

Data Science II Homework 3

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Split the dataset into two parts: training data (70%) and test data (30%).

```
library(tidyverse)
library(rsample)
library(caret)
library(earth)
library(MASS)
library(pROC)
library(pdp)
library(vip)
library(corrplot)

auto <- read.csv("auto.csv") %>%
  mutate(mpg_cat = factor(mpg_cat, levels = c("low", "high")),
         origin = factor(origin))

set.seed(2026)
data_split <- initial_split(auto, prop = 0.7, strata = mpg_cat)
train_data <- training(data_split)
test_data <- testing(data_split)

ctrl <- trainControl(method = "repeatedcv", number = 10, repeats = 5,
                     summaryFunction = twoClassSummary,
                     classProbs = TRUE)
```

(a) Logistic Regression

```
glm.fit <- glm(mpg_cat ~ ., data = train_data, family = binomial(link = "logit"))
summary(glm.fit)
```

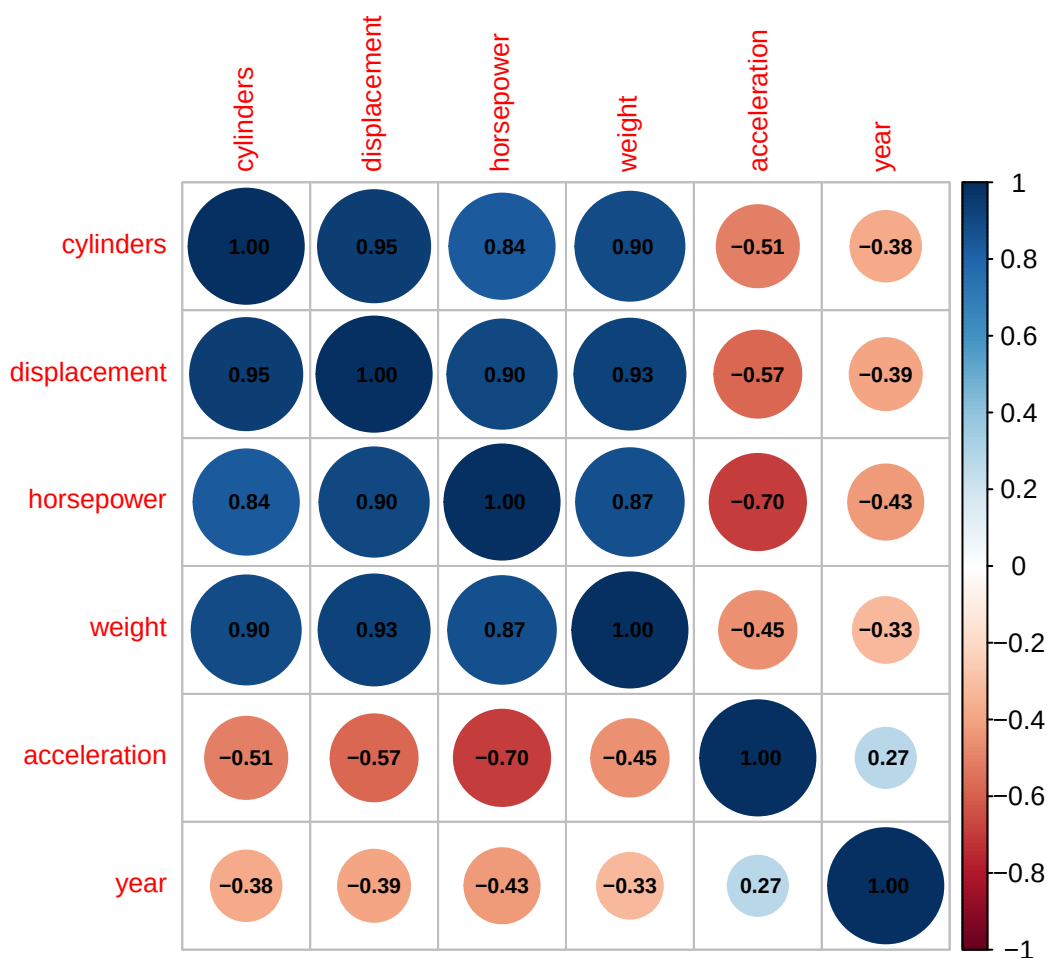
```
##
## Call:
## glm(formula = mpg_cat ~ ., family = binomial(link = "logit"),
##      data = train_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -29.689750   8.995316  -3.301 0.000965 ***
```

```
## cylinders      0.303958   0.564008   0.539 0.589939
## displacement  0.003788   0.018320   0.207 0.836184
## horsepower    -0.047100   0.034054  -1.383 0.166633
## weight        -0.006755   0.001956  -3.454 0.000552 ***
## acceleration  0.273092   0.206630   1.322 0.186285
## year          0.605390   0.128613   4.707 2.51e-06 ***
## origin2       1.759034   1.019903   1.725 0.084580 .
## origin3       1.642283   0.982850   1.671 0.094734 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 379.845  on 273  degrees of freedom
## Residual deviance:  89.978  on 265  degrees of freedom
## AIC: 107.98
##
## Number of Fisher Scoring iterations: 8
```

```
# Fit logistic regression using caret for later comparison
set.seed(2026)
model.glm <- train(mpg_cat ~ .,
                   data = train_data,
                   method = "glm",
                   metric = "ROC",
                   trControl = ctrl)
```

We examine the correlation structure among numeric predictors:

```
num_vars <- train_data[, sapply(train_data, is.numeric)]
corrplot(cor(num_vars), method = "circle", type = "full", addCoef.col = "black",
         tl.cex = 0.8, number.cex = 0.7)
```



Yes, there are redundant predictors. The correlation plot shows that `cylinders`, `displacement`, `horsepower`, and `weight` are highly correlated with one another. In the logistic regression summary, `cylinders`, `displacement`, `horsepower`, and `acceleration` have large p-values (> 0.05), indicating they do not contribute significant additional information beyond the other predictors. These variables are redundant due to multicollinearity — they carry overlapping information about engine size and vehicle performance.

(b) MARS Model

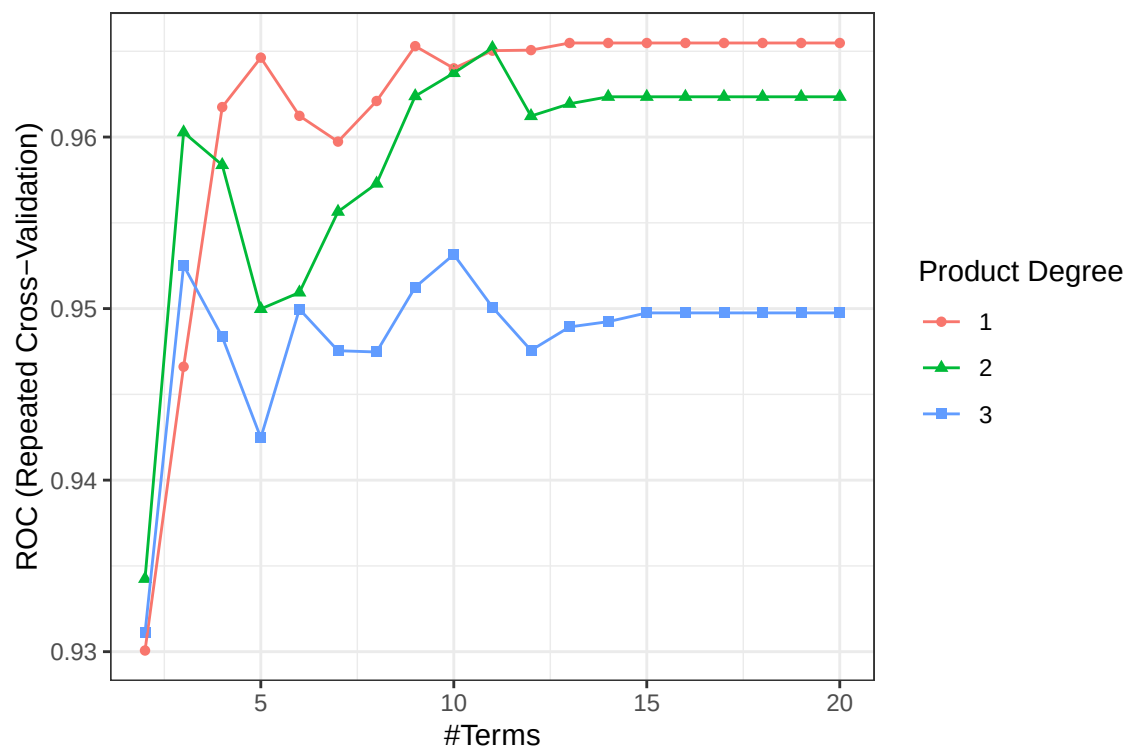
```

mars_grid <- expand.grid(degree = 1:3, nprune = 2:20)

set.seed(2026)
model.mars <- train(mpg_cat ~ .,
  data = train_data,
  method = "earth",
  tuneGrid = mars_grid,
  metric = "ROC",
  trControl = ctrl,
  glm = list(family = binomial))

ggplot(model.mars) + theme_bw()

```



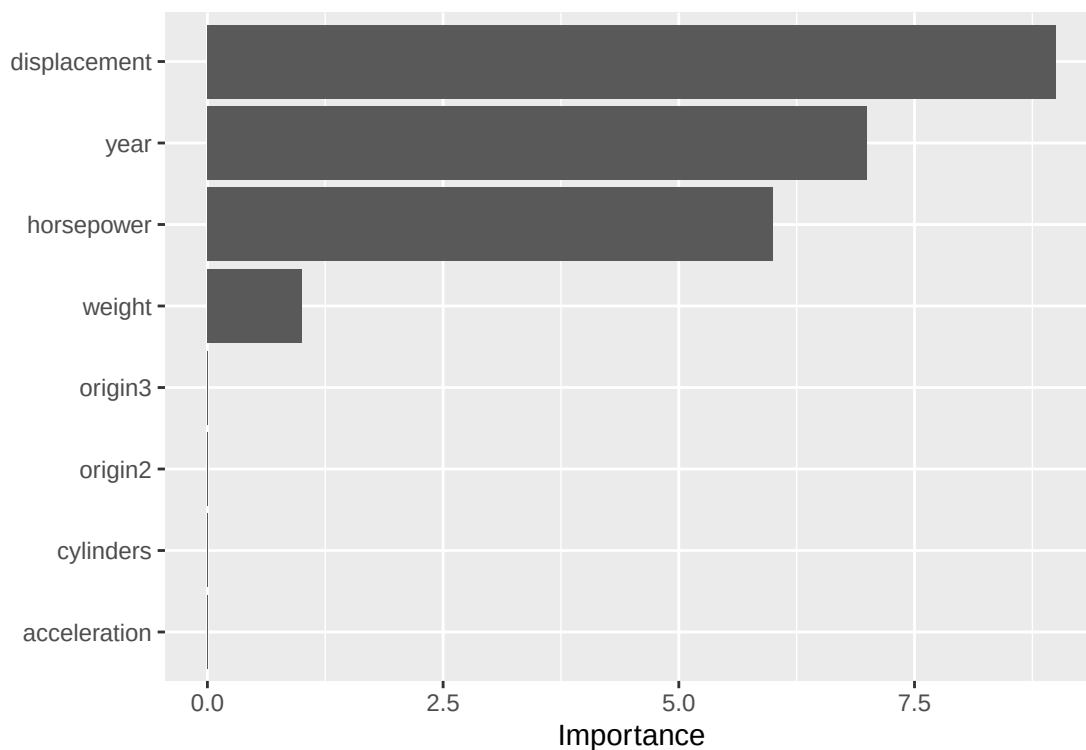
```
model.mars$bestTune
```

```
##      nprune degree
## 12      13      1
```

```
coef(model.mars$finalModel)
```

```
##      (Intercept)      h(year-78) h(horsepower-133) h(displacement-173)
##      0.671035800      1.006047154      -6.326174879      -0.318151894
## h(displacement-262) h(3155-weight) h(horsepower-80) h(displacement-200)
##      -0.702905504      0.002620313      -0.120769804      0.112490175
## h(displacement-140) h(displacement-250)
##      0.103968817      0.813237846
```

```
vip(model.mars$finalModel, type = "nsubsets")
```

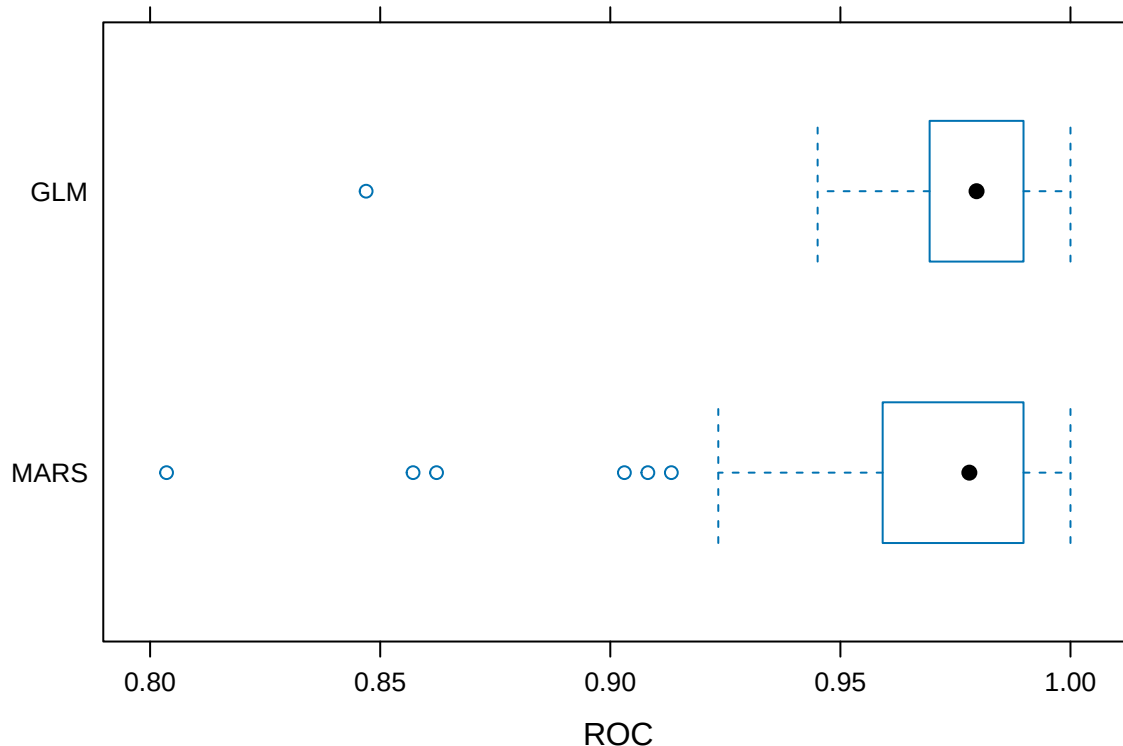


```
# Compare CV performance
```

```
res_ab <- resamples(list(GLM = model.glm, MARS = model.mars))
summary(res_ab)
```

```
##
## Call:
## summary.resamples(object = res_ab)
##
## Models: GLM, MARS
## Number of resamples: 50
##
## ROC
##           Min.   1st Qu.   Median     Mean   3rd Qu. Max. NA's
## GLM  0.8469388 0.9693878 0.9795918 0.9762523 0.9897959    1    0
## MARS 0.8035714 0.9597724 0.9780220 0.9654833 0.9897959    1    0
##
## Sens
##           Min.   1st Qu.   Median     Mean   3rd Qu. Max. NA's
## GLM  0.6428571 0.8571429 0.9230769 0.8967033 0.9285714    1    0
## MARS 0.6428571 0.8571429 0.9258242 0.8950549 0.9285714    1    0
##
## Spec
##           Min.   1st Qu.   Median     Mean   3rd Qu. Max. NA's
## GLM  0.7142857 0.8571429 0.9285714 0.9151648 0.9285714    1    0
## MARS 0.6428571 0.8736264 0.9285714 0.9238462 1.0000000    1    0
```

```
bwplot(res_ab, metric = "ROC")
```



```
# Compare test set performance
glm.pred.b <- predict(model.glm, newdata = test_data, type = "prob")[, 2]
mars.pred.b <- predict(model.mars, newdata = test_data, type = "prob")[, 2]
roc.glm.b <- roc(test_data$mpg_cat, glm.pred.b)
roc.mars.b <- roc(test_data$mpg_cat, mars.pred.b)
auc.glm.b <- roc.glm.b$auc[1]
auc.mars.b <- roc.mars.b$auc[1]
cat("Test AUC - GLM:", round(auc.glm.b, 4), "\n")
```

```
## Test AUC - GLM: 0.9555
```

```
cat("Test AUC - MARS:", round(auc.mars.b, 4), "\n")
```

```
## Test AUC - MARS: 0.9681
```

Based on both the cross-validation and test set AUC, the MARS model achieves a comparable (or slightly higher) AUC relative to logistic regression.

(c) Linear Discriminant Analysis

```

set.seed(2026)
model.lda <- train(mpg_cat ~ .,
                   data = train_data,
                   method = "lda",
                   metric = "ROC",
                   trControl = ctrl)
model.lda$finalModel

## Call:
## lda(x, grouping = y)
##
## Prior probabilities of groups:
##   low high
## 0.5  0.5
##
## Group means:
##      cylinders displacement horsepower   weight acceleration   year   origin2
## low   6.832117      275.0949  130.62044 3622.708      14.39051 74.27007 0.0729927
## high  4.197080      116.4708   78.24818 2336.752      16.59489 77.65693 0.2408759
##      origin3
## low  0.02919708
## high 0.38686131
##
## Coefficients of linear discriminants:
##              LD1
## cylinders    -0.364786740
## displacement  0.001091106
## horsepower    0.010579890
## weight       -0.001370760
## acceleration  0.061376381
## year         0.126273484
## origin2      0.374880337
## origin3      0.462938751

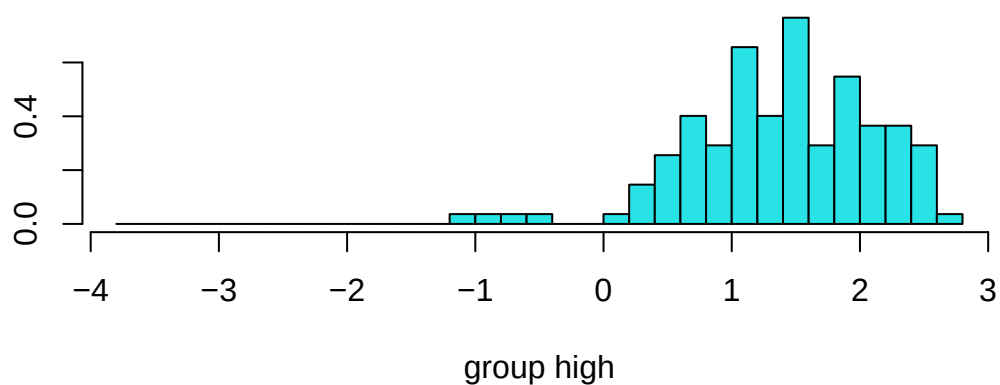
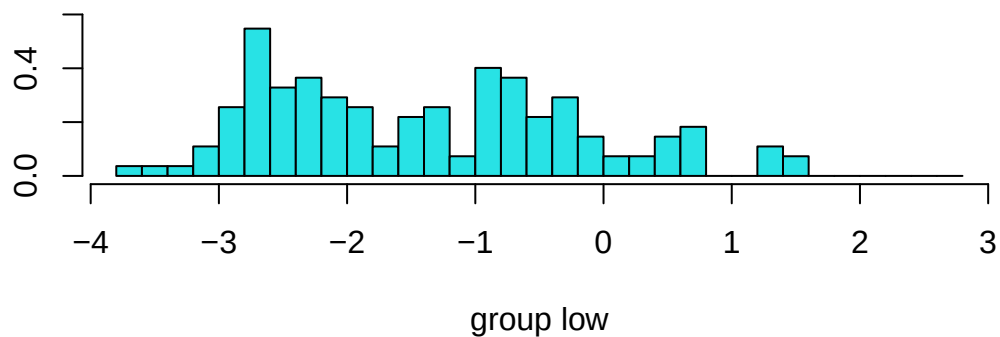
```

Since there are $K = 2$ classes, there is only one linear discriminant. We plot the histogram of the discriminant scores for each class:

```

x_train <- model.matrix(mpg_cat ~ ., data = train_data)[, -1]
lda.scores <- predict(model.lda$finalModel, x_train)$x[, 1]
ldahist(lda.scores, g = train_data$mpg_cat)

```



(d) Model Comparison and ROC Curve

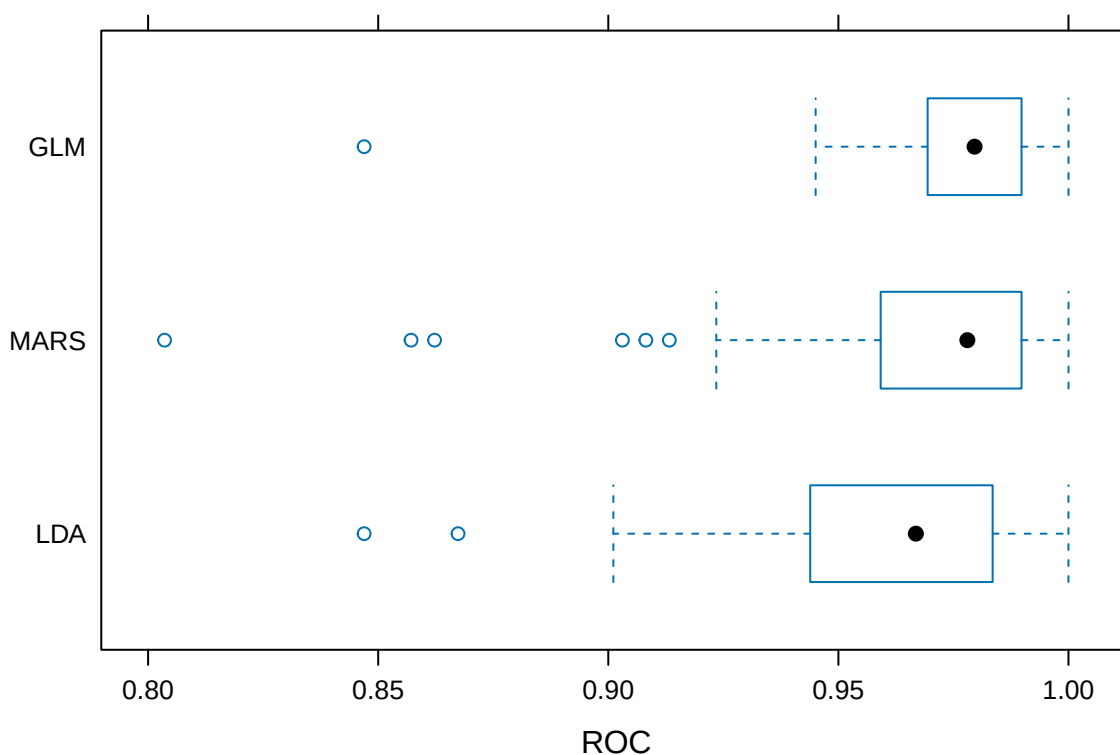
```
# Compare all three models via CV
res_all <- resamples(list(GLM = model.glm, MARS = model.mars, LDA = model.lda))
summary(res_all)
```

```
##
## Call:
## summary.resamples(object = res_all)
##
## Models: GLM, MARS, LDA
## Number of resamples: 50
##
## ROC
```



```
##           Min.    1st Qu.    Median      Mean    3rd Qu. Max. NA's
## GLM  0.8469388 0.9693878 0.9795918 0.9762523 0.9897959    1    0
## MARS 0.8035714 0.9597724 0.9780220 0.9654833 0.9897959    1    0
## LDA  0.8469388 0.9441719 0.9668367 0.9619044 0.9835165    1    0
##
## Sens
##           Min.    1st Qu.    Median      Mean    3rd Qu. Max. NA's
## GLM  0.6428571 0.8571429 0.9230769 0.8967033 0.9285714    1    0
## MARS 0.6428571 0.8571429 0.9258242 0.8950549 0.9285714    1    0
## LDA  0.5714286 0.7857143 0.8571429 0.8556044 0.9285714    1    0
##
## Spec
##           Min.    1st Qu.    Median      Mean    3rd Qu. Max. NA's
## GLM  0.7142857 0.8571429 0.9285714 0.9151648 0.9285714    1    0
## MARS 0.6428571 0.8736264 0.9285714 0.9238462 1.0000000    1    0
## LDA  0.8461538 0.9285714 1.0000000 0.9631868 1.0000000    1    0
```

```
bwplot(res_all, metric = "ROC")
```



```
# Test set predictions
glm.pred <- predict(model.glm, newdata = test_data, type = "prob")[, 2]
mars.pred <- predict(model.mars, newdata = test_data, type = "prob")[, 2]
lda.pred <- predict(model.lda, newdata = test_data, type = "prob")[, 2]

# ROC curves
```

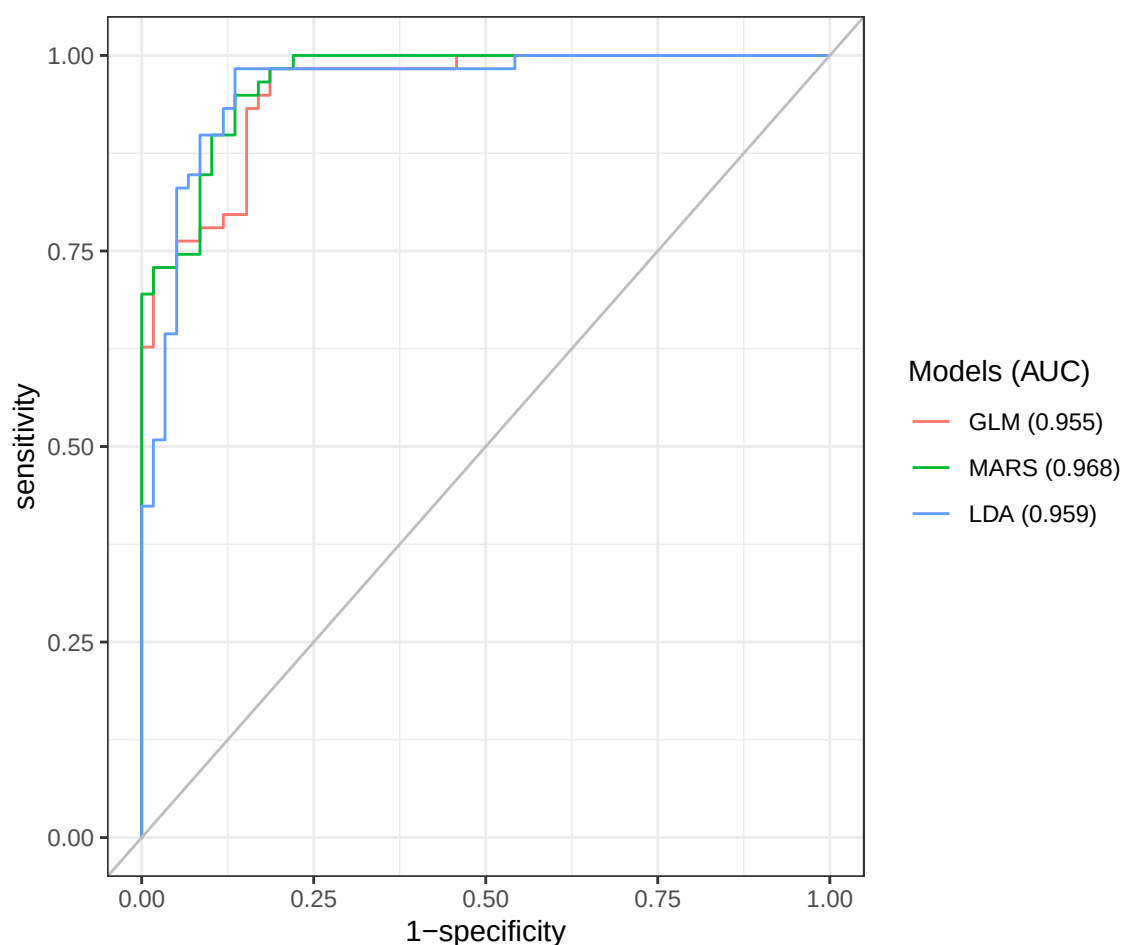
```

roc.glm <- roc(test_data$mpg_cat, glm.pred)
roc.mars <- roc(test_data$mpg_cat, mars.pred)
roc.lda <- roc(test_data$mpg_cat, lda.pred)

auc <- c(roc.glm$auc[1], roc.mars$auc[1], roc.lda$auc[1])
modelNames <- c("GLM", "MARS", "LDA")

ggroc(list(roc.glm, roc.mars, roc.lda), legacy.axes = TRUE) +
  scale_color_discrete(labels = paste0(modelNames, " (", round(auc, 3), ")"),
    name = "Models (AUC)") +
  geom_abline(intercept = 0, slope = 1, color = "grey") +
  theme_bw()

```



```

cat("Selected model:", modelNames[which.max(auc)],
    "with test AUC:", round(max(auc), 4), "\n")

```

```
## Selected model: MARS with test AUC: 0.9681
```

```

# Number of original predictors used by each model
n_glm <- ncol(model.glm$finalModel$model) - 1
n_mars <- length(evimp(model.mars$finalModel)[, 1])

```

```
n_lda <- nrow(model.lda$finalModel$scaling)
cat("\nNumber of predictors used:\n")
```

```
##
## Number of predictors used:
```

```
cat("GLM:", n_glm, "\nMARS:", n_mars, "\nLDA:", n_lda, "\n")
```

```
## GLM: 8
## MARS: 4
## LDA: 8
```

The three models have comparable test AUC values, but differ in complexity. GLM and LDA use all predictors, while MARS performs automatic variable selection and retains fewer terms. Since the MARS model achieves the highest test AUC with a simpler model structure, it is preferred on the grounds of both predictive accuracy and parsimony. Therefore, we select MARS as our final model.

(e) Confusion Matrix

We use Youden's J statistic to select the optimal classification threshold, which maximizes the sum of sensitivity and specificity.

```
threshold <- coords(roc.mars, "best", ret = "threshold",
                    best.method = "youden")$threshold
cat("Optimal threshold (Youden's J):", round(threshold, 3), "\n")
```

```
## Optimal threshold (Youden's J): 0.458
```

```
test.pred <- factor(ifelse(mars.pred > threshold, "high", "low"),
                    levels = c("low", "high"))
```

```
confusionMatrix(data = test.pred,
                 reference = test_data$mpg_cat,
                 positive = "high")
```

```
## Confusion Matrix and Statistics
```

```
##
```

```
##           Reference
```

```
## Prediction low high
```

```
##          low   51    3
```

```
##          high    8   56
```

```
##
```

```
##           Accuracy : 0.9068
```

```
##           95% CI : (0.8393, 0.9525)
```

```
## No Information Rate : 0.5
```

```
## P-Value [Acc > NIR] : <2e-16
```

```
##
```

```
##           Kappa : 0.8136
```

```
##
```

```
## McNemar's Test P-Value : 0.2278
##
##           Sensitivity : 0.9492
##           Specificity : 0.8644
##           Pos Pred Value : 0.8750
##           Neg Pred Value : 0.9444
##           Prevalence : 0.5000
##           Detection Rate : 0.4746
##           Detection Prevalence : 0.5424
##           Balanced Accuracy : 0.9068
##
##           'Positive' Class : high
##
```

The confusion matrix indicates that our selected MARS model has strong predictive power. By applying the Youden-derived threshold, we achieved a high rate of correct classifications for both 'high' and 'low' mpg cars. The minimal number of misclassifications suggests the model generalizes well to new data, and the balanced accuracy proves it is reliable across both categories.